

Department: Computer Engineering & Applications

Programme: B.Tech

Branch: CSE

Subject Name: Application Build & Release
Automation

Subject Code: BCSE0021

Assignment Instructions

- Submit all answers in 3 A4-size pages (handwritten).
- Write clearly and organize your answers properly, using headings where necessary.
- Ensure that all questions are attempted.
- Upload the assignment on or before the given deadline.

Q1. Your college's academic cell currently manages all lab manuals using a Centralized Version Control System (CVCS) hosted on a single department server. During a semester-end rush, the server crashes for two days, and no faculty member can access, edit, or commit changes to any manual. (a) Explain why this failure would NOT have completely stopped work if the department had instead used a Distributed VCS (DVCS) like Git. (b) Identify and briefly describe two key characteristics of DVCS that address this specific problem.

Q2. A student, John, is working on a Git-tracked project. She modifies three files: `config.py`, `utils.py`, and `main.py`. She wants to commit only the changes in `config.py` and `utils.py` for now, keeping `main.py`'s changes for a later, separate commit. (a) Explain, step by step, how Git's three-tree architecture (Working Directory, Staging Area, Repository) allows John to do this. (b) Write the exact Git commands she should run to achieve this, and explain what happens internally at each step.

Q3. A team of four developers is collaborating on a shared GitHub repository for a college project. One developer, Aman, has just finished a new feature on his local machine and wants to make it available to the rest of the team, while also ensuring he has everyone else's latest changes before he starts. Describe the complete Git workflow Aman should follow, from making local changes to sharing them with the team, including the specific commands involved and the purpose of each.

Q4. Your team is developing a student result-management web application. The main branch always contains the stable, deployed version. A new requirement comes in: add an SMS notification feature, which will take about two weeks to build and test. (a) Explain why working directly on the main branch would be risky in this scenario. (b) Describe a suitable branching strategy for this situation, including the commands to create, switch to, and later merge the feature branch.

Q5. A new intern joins your DevOps team on their first day. They are given access to an existing GitHub repository containing the project codebase, and are asked to set up the project locally, make a small fix, and share it back with the team — while also staying updated with any changes teammates push during the day. Walk through the sequence of remote operations (with commands) the intern should perform, explaining the purpose of each (clone, pull, push) and what could go wrong if they skip the pull step before pushing.

Q6. Two developers, John and Katherin, both edit the same function in `database.py` on their own local branches and then attempt to merge their branches into main. Git reports a merge conflict. (a) Explain why Git is unable to merge these changes automatically. (b) Describe, step by step, how John should identify, resolve, and finalize the conflict, including what the conflict markers in the file mean.

Q7. Your college's placement cell wants to deploy a Python-based resume-screening application. The system admin suggests using a Virtual Machine, but your DevOps instructor recommends Docker containers instead. (a) Compare Virtual Machines and Docker containers on the basis of resource usage, startup time, and isolation. (b) Justify, using this scenario, why Docker containers would be a more efficient choice here.

Q8. When you run the command ``docker run -d -p 080:80 nginx`` on your laptop for the first time, several components of Docker's architecture work together behind the scenes to get the Nginx web server running. Explain the role of the Docker Client, Docker Daemon, and Docker Engine in executing this single command, describing what happens at each stage from the moment you press Enter to the container running.

Q9. You are asked to containerize a simple Node.js web application for your college's mini-project. The application listens on port 3000, requires installing dependencies from `package.json`, and should start automatically when the container runs. Write a suitable Dockerfile for this application, and briefly explain the purpose of each instruction you include.

Q10. You are containerizing a two-part application for a college project: a MySQL database container and a web application container that connects to it. You are concerned about two things — (1) losing all database data if the MySQL container is removed, and (2) ensuring the web app container can communicate with the database container by name. (a) Explain how Docker Volumes would solve the data-loss concern, and name the type of volume best suited here. (b) Explain how Docker Networking (and network types) would solve the communication concern, recommending a specific Docker network type with justification.